

Activity 5: EBQ and Kanban Control Design

Level: Intermediate Duration: 75 minutes Mapping: K7 A3 Tools: Calculation worksheet; calculator; checklist

Goal

Calculate EBQ and kanban quantity, test assumptions and design a signal-and-response rule.

Learner output

EBQ calculation, kanban quantity, sensitivity table, card/bin rule and review trigger.

Files in this folder

- README.md / README.pdf - activity guide and detailed procedure
- scenario.md / scenario.pdf - case facts and questions
- worksheet.md / worksheet.pdf - structured response template
- evidence-checklist.md / evidence-checklist.pdf - acceptance record
- data.csv - mock dataset when analysis is required

Detailed procedure

1. Record annual demand D , setup cost S , holding cost H and production rate P .
2. Confirm P is greater than D and that units and time bases match.
3. Calculate EBQ using the finite-production formula.
4. Recalculate EBQ after a 50% setup-cost reduction and explain the lean implication.
5. Record daily demand, lead time, safety factor and container quantity.
6. Calculate kanban quantity and round up to a whole container.
7. Test lead time and safety-factor sensitivity.
8. Write the physical or electronic signal rule, owner and abnormal-condition response.

Verification

Complete every acceptance item in evidence-checklist.md. Submit the worksheet plus calculations, exported tool output or screenshots named in the worksheet.

Safety, privacy and responsible AI

Use only the supplied scenario data unless your organisation has authorised another dataset. Remove personal, confidential and export-controlled information. AI outputs are suggestions until verified by the named human decision owner.